

# Qatar-1b

R-2617

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-1\_250920 · created 2026-08-28T23:07:09+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460938.6393 +/- 0.0031 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.126 +/- 0.054                 |
| Transit depth (Rp/Rs)^2                           | 1.58 +/- 1.35 [%]               |
| Orbital Inclination (inc)                         | 83.55 +/- 2.89                  |
| Airmass coefficient 1 (a1)                        | 0.028 +/- 0.014                 |
| Airmass coefficient 2 (a2)                        | 2.95 +/- 1.69                   |
| Scatter in the residuals of the lightcurve fit is | 51.45 %                         |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 2.17                            |
| Optimal Annulus                                   | 4.5                             |
| Transit Duration (day)                            | 0.044 +/- 0.018                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.126 ± 0.054 vs 0.1463 (deviation 0.3σ) |
| Duration fit vs published | 0.044 d vs 0.0692 d (deviation 1.1σ)     |
| Mid-time O–C              | -6.9 min                                 |
| Depth SNR                 | 1.8                                      |
| Residual scatter          | 51.45 %                                  |

### Light curve

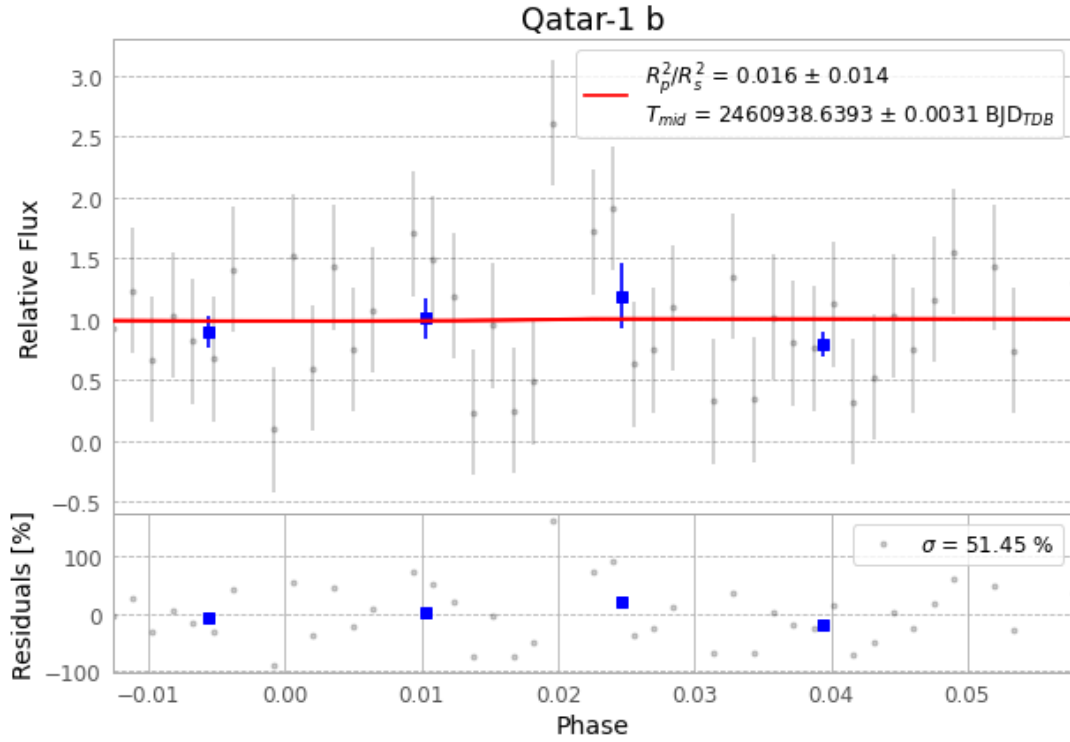


Figure 1 — FinalLightCurve\_Qatar-1 b\_2025-09-19.png (SHA-256 9fcba48b09dae87e...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [432, 79], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 aff1271aeda300d5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

49 FITS frames (32 MB), Qatar-1250920025116.FITS → Qatar-1250920051515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-250920.log (dede2323d455...), FinalLightCurve\_Qatar-1 b\_2025-09-19.png (9fcba48b09da...), FinalLightCurve\_Qatar-1 b\_2025-09-19.csv (0e0a2e0ee29e...)

## Compute resources

Wall time 154.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 23:07Z.